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Statistics Module Documentation

SupplierService

Section	Description
Overview	Service layer for managing suppliers with data integrity checks, attachment handling, and relationship tracking.
Dependencies	- IUnitOfWork – Repository & transactions - IWebHostEnvironment – File attachment handling

SupplierService Operations

Method	Purpose	Key Notes
GetAllAsync()	Retrieve all suppliers	- Uses AsNoTracking() - Dynamically sets HasRelatedData - Checks Expenses, Receipts, PurchaseOrders
GetByIdAsync(int id)	Fetch supplier by ID	Returns nullable Supplier entity
AddAsync(Supplier)	Create new supplier	Returns generated supplier ID
UpdateAsync(Supplier)	Update existing supplier	Includes null safety checks
DeleteAsync(int id)	Delete supplier	Prevents deletion if entity not found

Relationship Integrity Logic

Feature	Implementation
HasRelatedData	<code>context.Expenses.Any(e => e.SupplierId == s.Id) context.Receipts.Any(r => r.SupplierId == s.Id) context.PurchaseOrders.Any(po => po.SupplierId == s.Id)</code>

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Section	Description
Overview	Provides system-wide analytics across members, courses, activities, departments, and demographics.
Dependencies	- IMemberService - IUnitOfWork - IHttpContextAccessor (Localization)

Statistics Operations

Method	Purpose	Output
GetStatisticsAsync()	Collect and calculate all analytics	StatisticsDTO

Statistical Dimensions

Category	Description
Subscriptions	Activities, courses, department participation
Demographics	Age groups, gender, residents vs citizens
Departments	Course distribution and engagement rates

Division Safety Logic

Scenario	Handling
Zero Members	<code>if (allMembersCount == 0) { allMembersCount = 1; TotalInCourses = 1; TotalInActivities = 1; }</code>

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Statistics ViewModels

ViewModel	Purpose	Main Properties
StatisticsVM	Main container	TotalInActivities, TotalInCourses, All Stats Lists
AgeGroupStatVM	Age distribution	Label, Percentage, Count
CourseCategoryStatVM	Department participation	Label, Percentage, Count
GenderStatVM	Gender ratio	Label, Percentage, Count
ResidentsAndCitizensStatVM	Residency analytics	Label, Percentage, Count

Statistics DTOs

DTO	Purpose	Properties
StatisticsDTO	Transfer analytics data	TotalInActivities, TotalInCourses, Stats Lists
AgeGroupStat	Age data transfer	Label, Percentage, Count
CourseCategoryStat	Department data	Label, Percentage, Count
GenderStat	Gender data	Label, Percentage, Count
ResidentsAndCitizensStat	Residency data	Label, Percentage, Count

Business Intelligence Value

Area	Impact
Decision Making	Data-driven planning and optimization
Reporting	Printable and dashboard-ready analytics
Performance Tracking	Department and demographic insights